

Matplotlib Assignment

```
Matplotlib

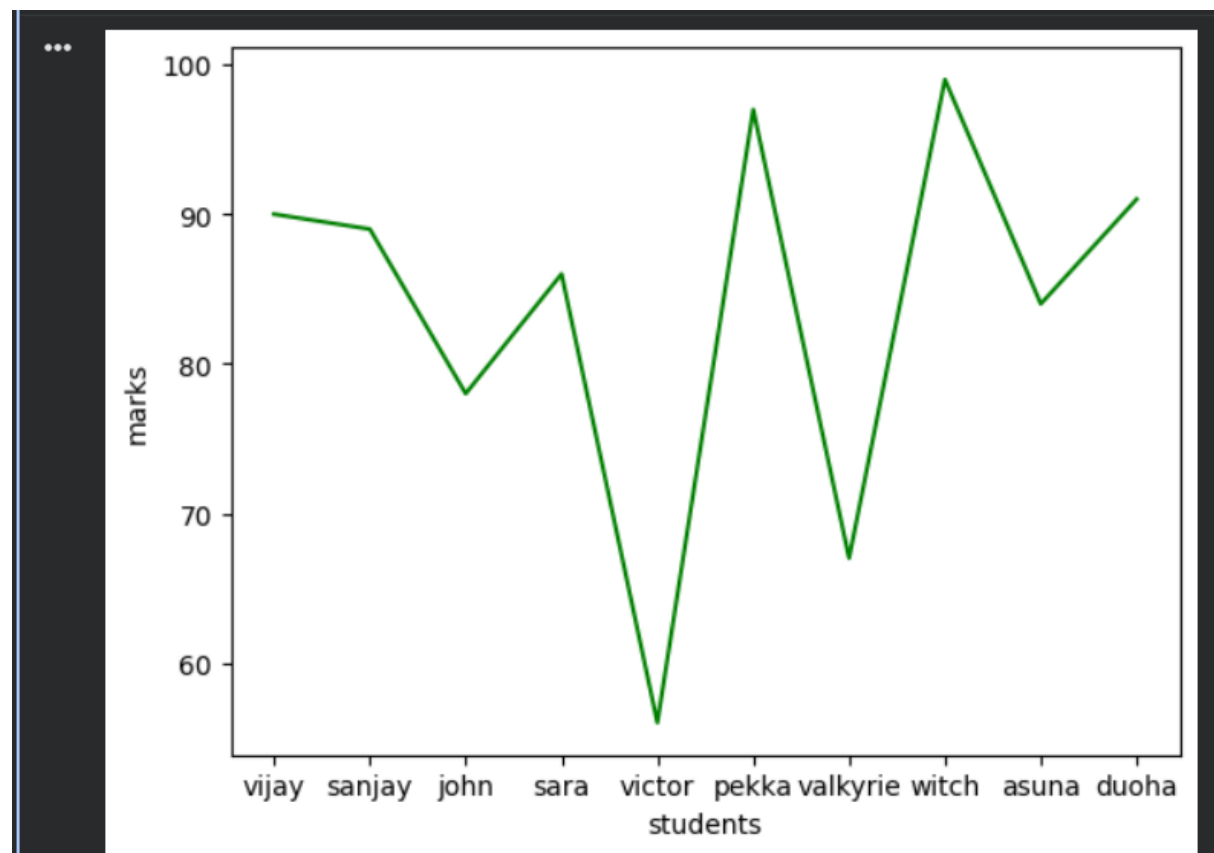
2] pip install matplotlib
✓ 5s

*** Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

1.

1.create a line graph showing marks of 10 students

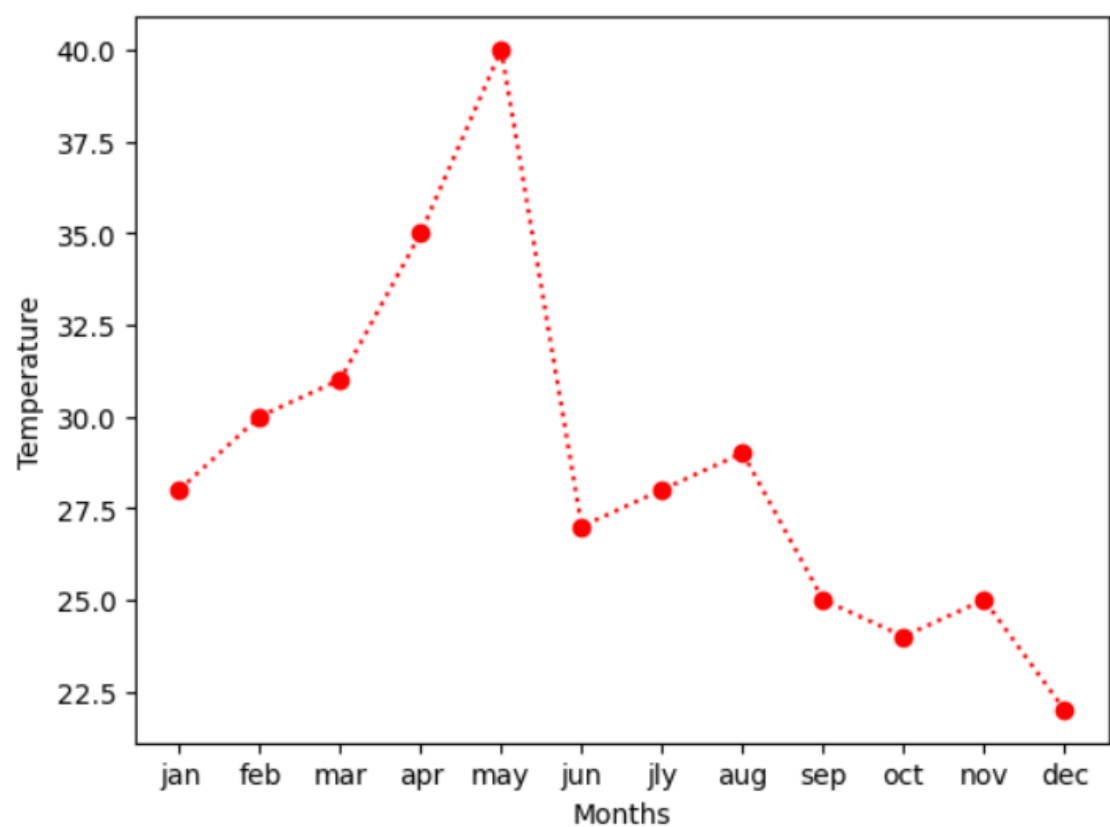
```
import matplotlib.pyplot as plt
students=["vijay","sanjay","john","sara","victor","pekka","valkyrie","witch","asuna","duoha"]
marks=[90,89,78,86,56,97,67,99,84,91]
plt.plot(students,marks,color="green")
plt.xlabel("students")
plt.ylabel("marks")
plt.show()
```



2.

2.plot monthly temperature

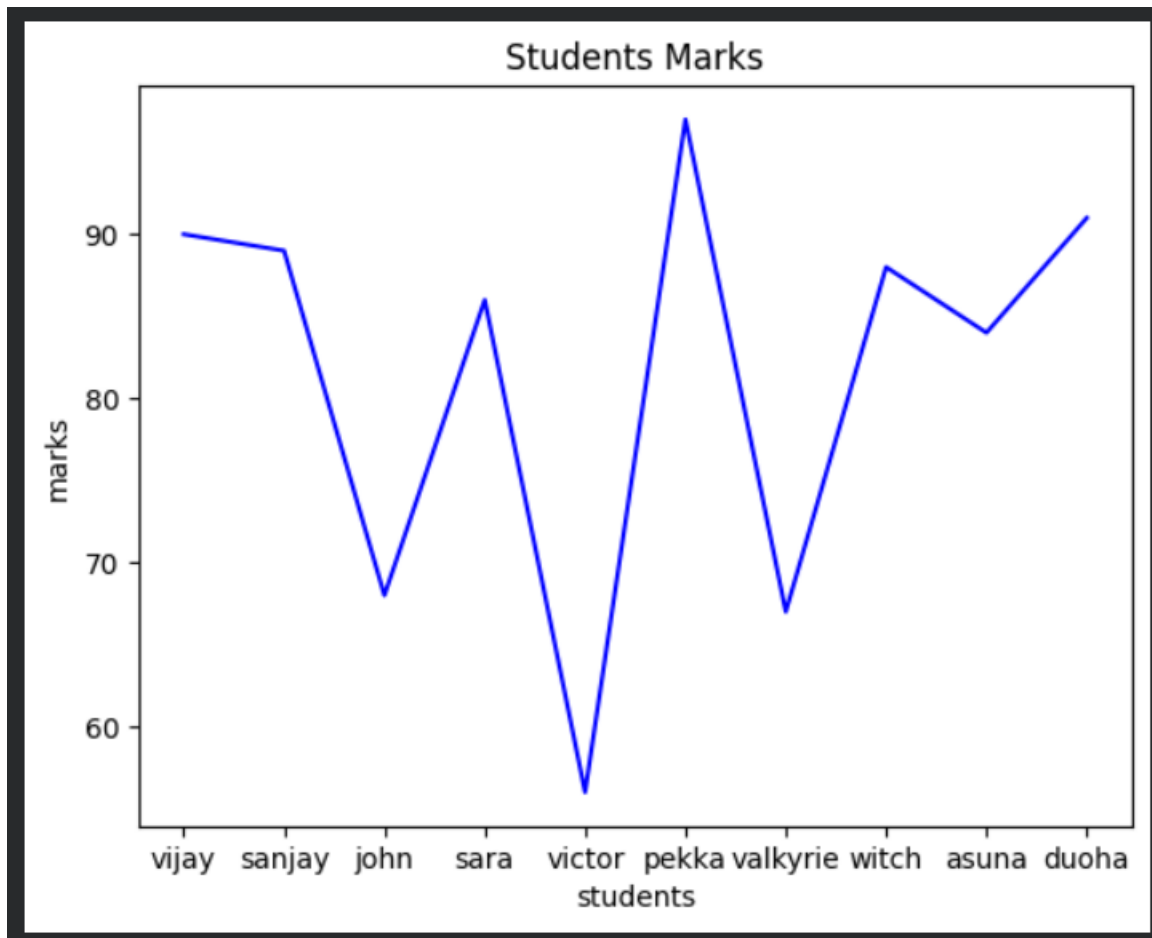
```
import matplotlib.pyplot as plt
month=["jan","feb","mar","apr","may","jun","jly","aug","sep","oct","nov","dec"]
temp=[28,30,31,35,40,27,28,29,25,24,25,22]
plt.plot(month,temp,color="red",linestyle=":",marker="o")
plt.xlabel("Months")
plt.ylabel("Temperature")
plt.show()
```



3.

3.add title,xlabel,ylabel

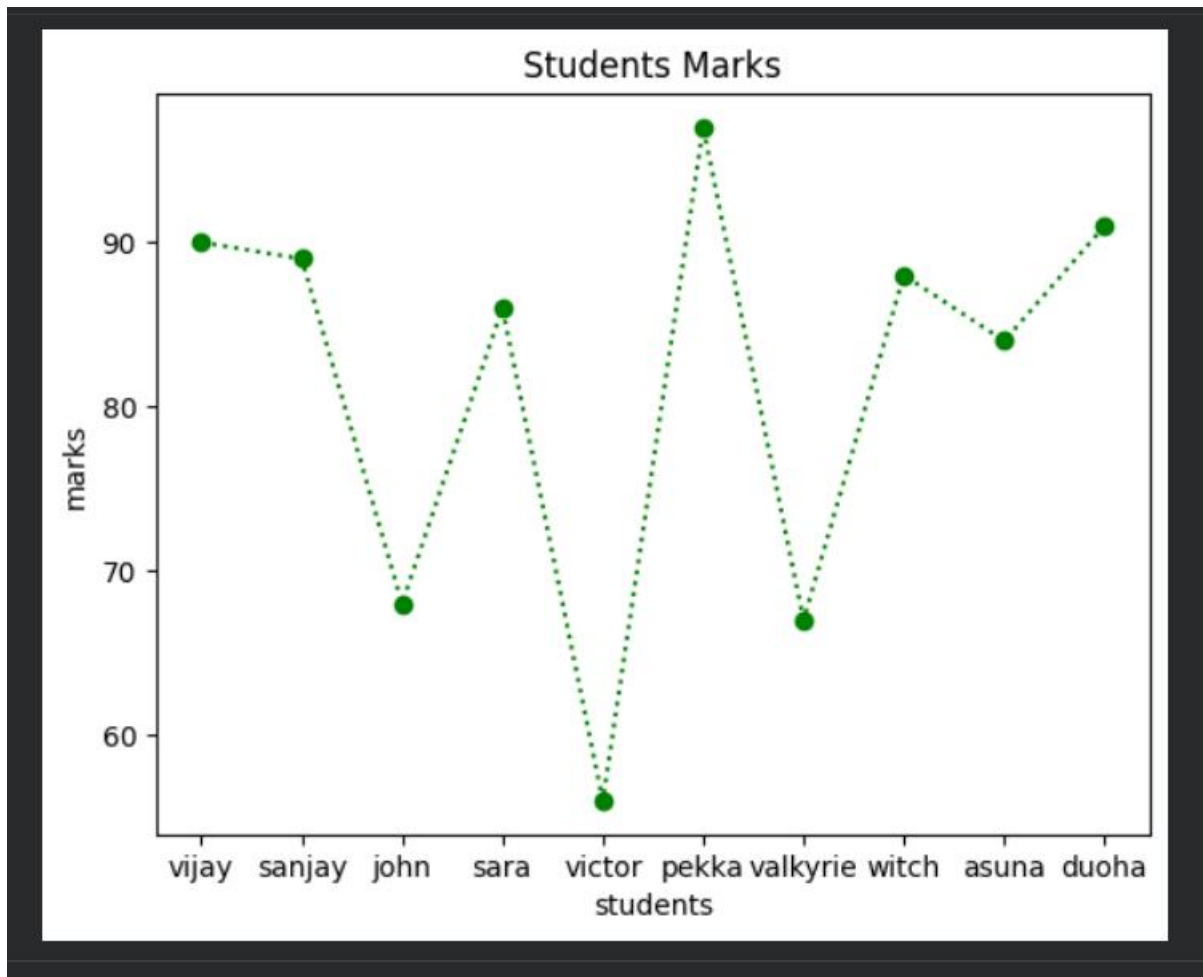
```
import matplotlib.pyplot as plt
students=["vijay","sanjay","john","sara","victor","pekka","valkyrie","witch","asuna","duoha"]
marks=[90,89,68,86,56,97,67,88,84,91]
plt.plot(students,marks,color="blue")
plt.title("Students Marks")
plt.xlabel("students")
plt.ylabel("marks")
plt.show()
```



4.

4.change color,marker and linestyle

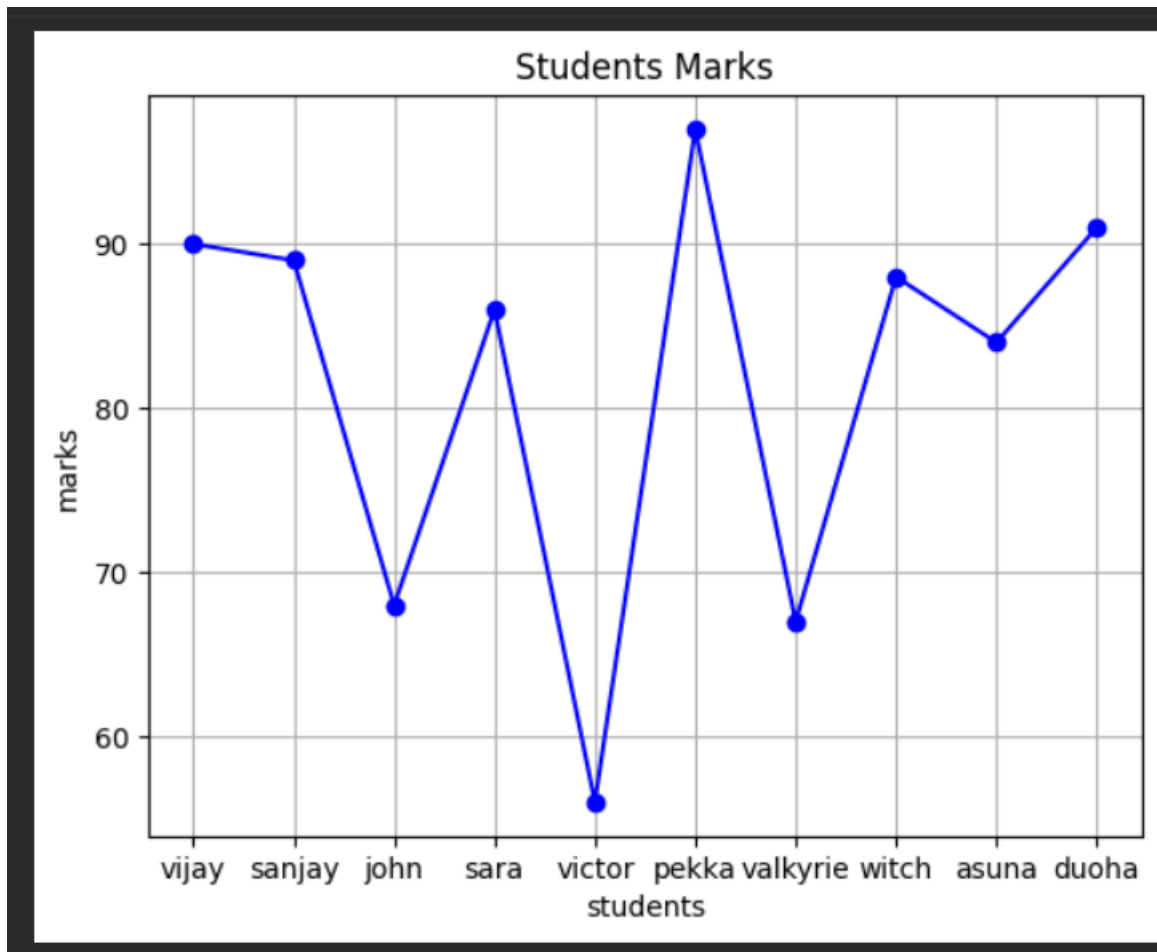
```
plt.plot(students,marks,color="green",marker="o",linestyle=":")  
plt.title("Students Marks")  
plt.xlabel("students")  
plt.ylabel("marks")  
plt.show()
```



5.

5.display graph and grid graph

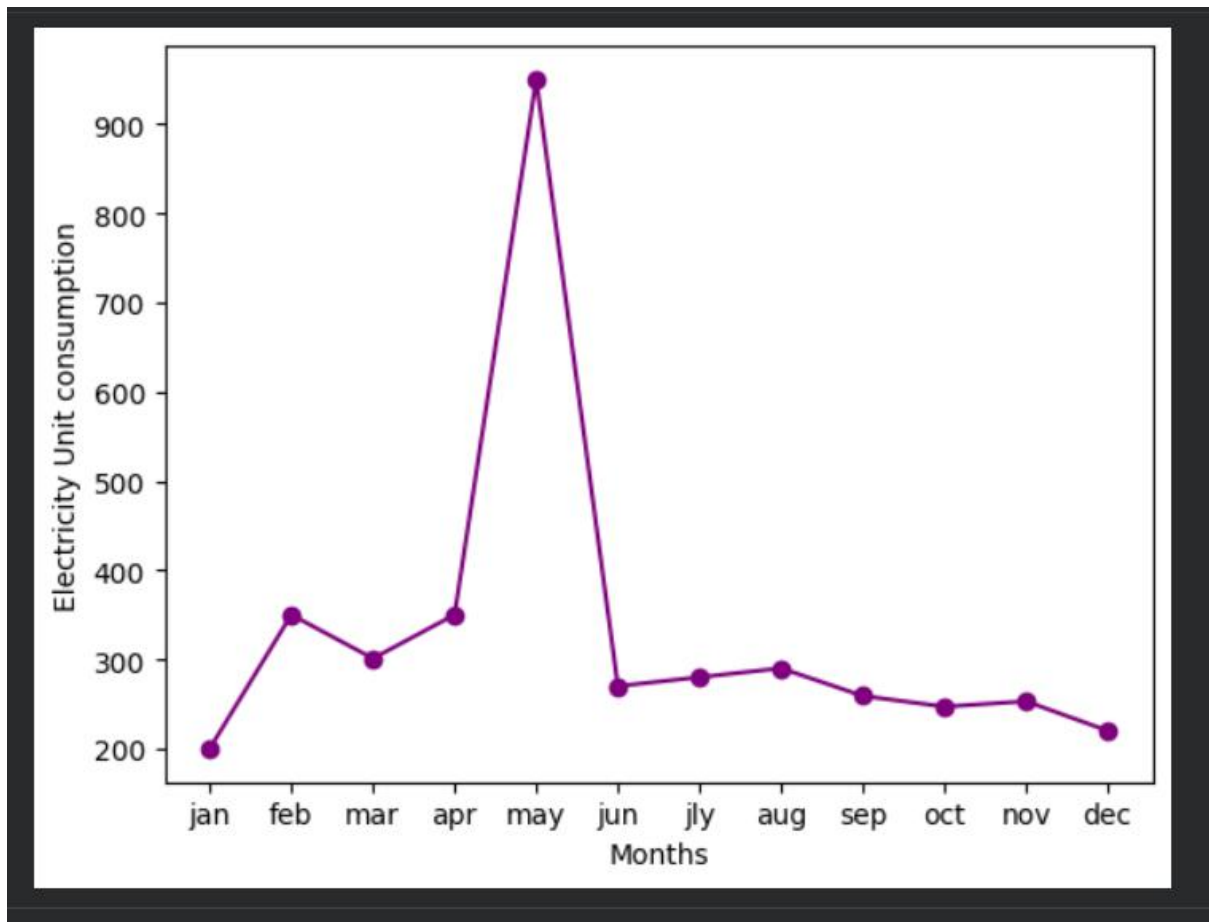
```
plt.plot(students,marks,color="blue",marker="o")  
plt.grid()  
plt.title("Students Marks")  
plt.xlabel("students")  
plt.ylabel("marks")  
plt.show()
```



6.

6.create line plot showing monthly electricity consumption

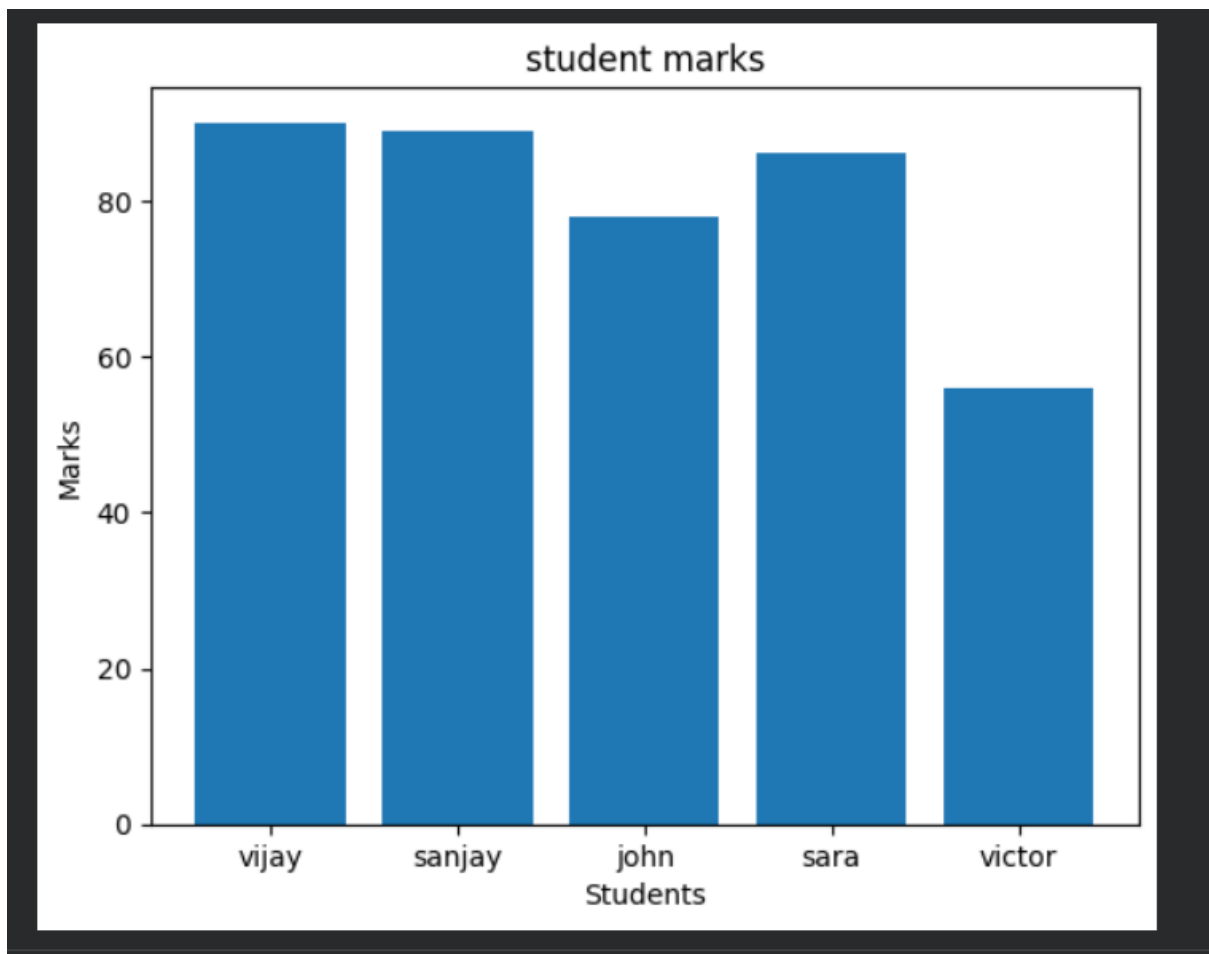
```
import matplotlib.pyplot as plt
month = ["jan", "feb", "mar", "apr", "may", "jun", "jly", "aug", "sep", "oct", "nov", "dec"]
elec = [200, 350, 301, 350, 950, 270, 280, 290, 259, 247, 253, 220]
plt.plot(month, elec, color="purple", marker="o")
plt.xlabel("Months")
plt.ylabel("Electricity Unit consumption")
plt.show()
```



7.

7.create a bar chart comparing five students marks

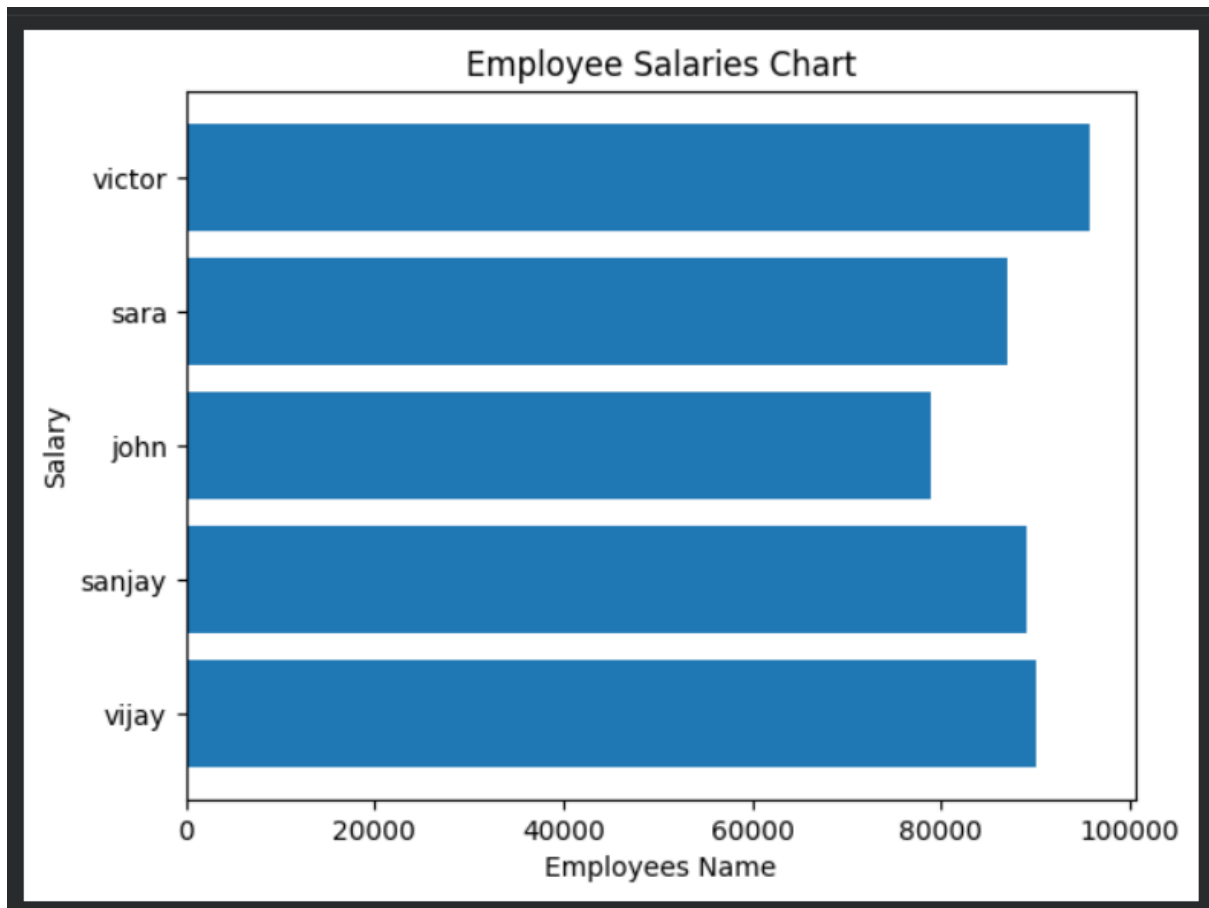
```
import matplotlib.pyplot as plt
students=["vijay","sanjay","john","sara","victor"]
marks=[90,89,78,86,56]
plt.bar(students,marks)
plt.title("student marks")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```



8.

8.create a horizontal barchart showing employee salaries

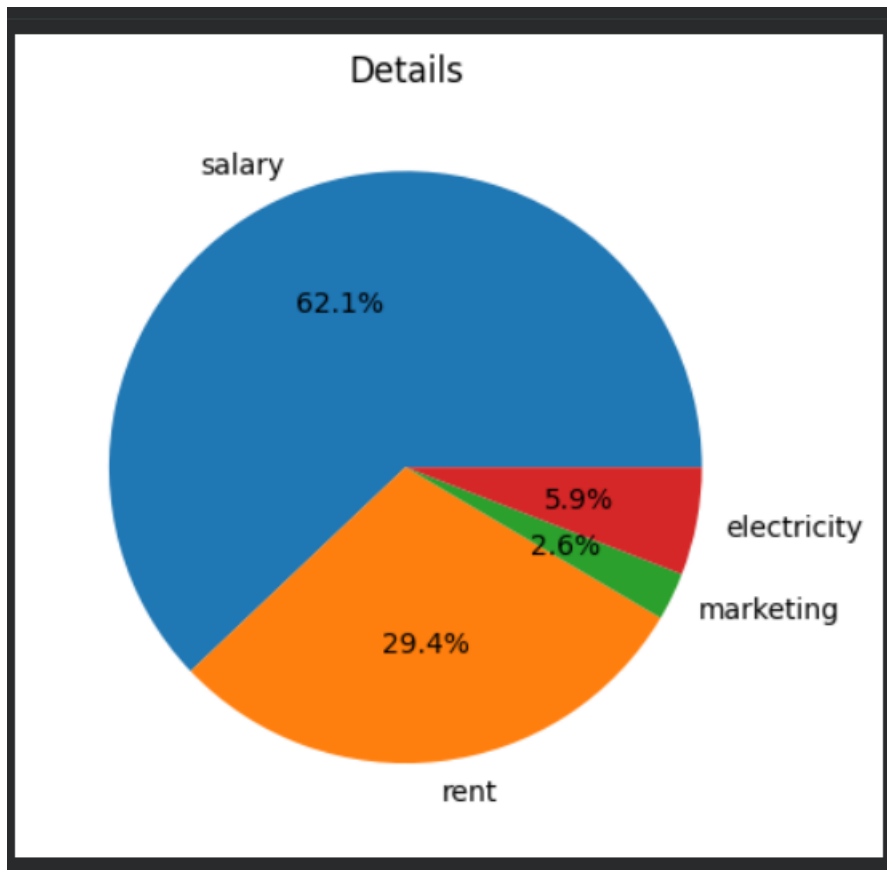
```
import matplotlib.pyplot as plt
employee = ["vijay", "sanjay", "john", "sara", "victor"]
salary = [90000, 89000, 78966, 86974, 95748]
plt.barh(employee, salary)
plt.title("Employee Salaries Chart")
plt.xlabel("Employees Name")
plt.ylabel("Salary")
plt.show()
```



9.

9.create a pie chart showing company expenses like , salary,rent,marketing,electricity

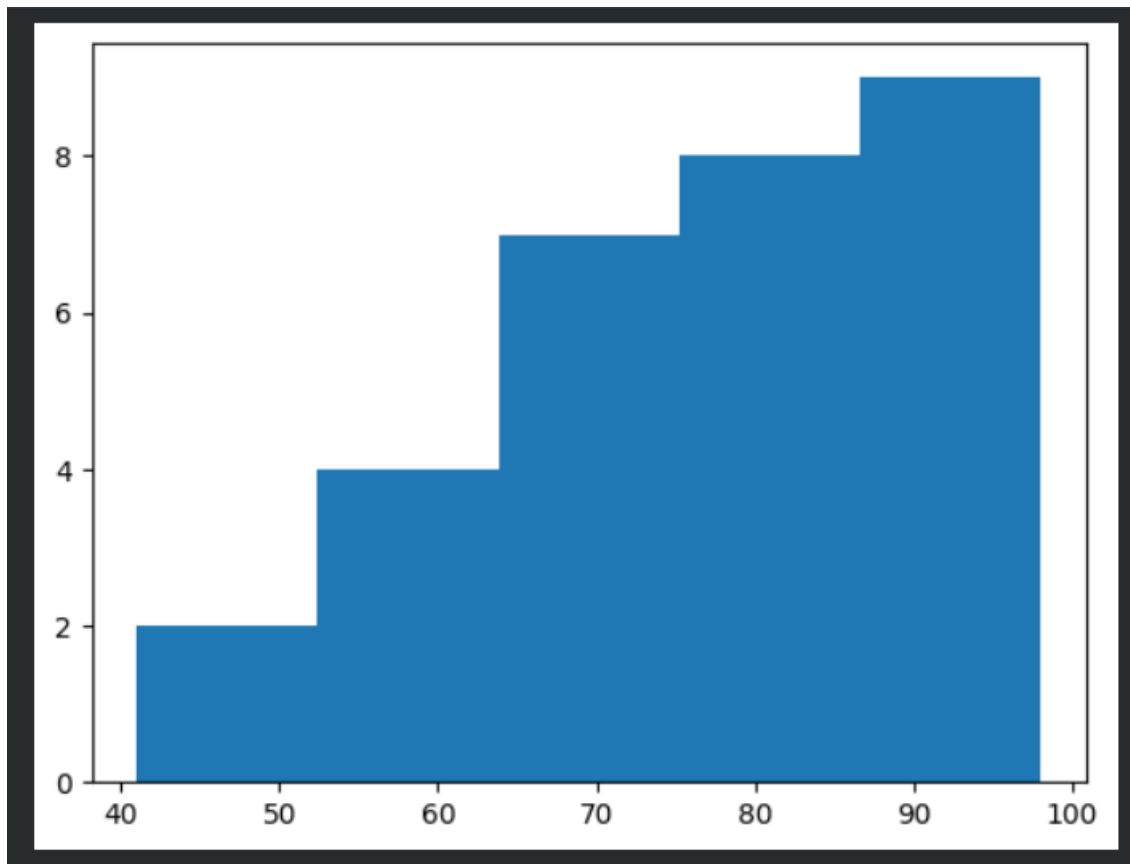
```
import matplotlib.pyplot as plt
categories=["salary","rent","marketing","electricity"]
company=[95000.0,45000.0,4000.0,9000.0]
plt.pie(company,labels=categories,autopct="%1.1f%%")
plt.title("Details")
plt.show()
```

10.

10.create a histogram for 30 students

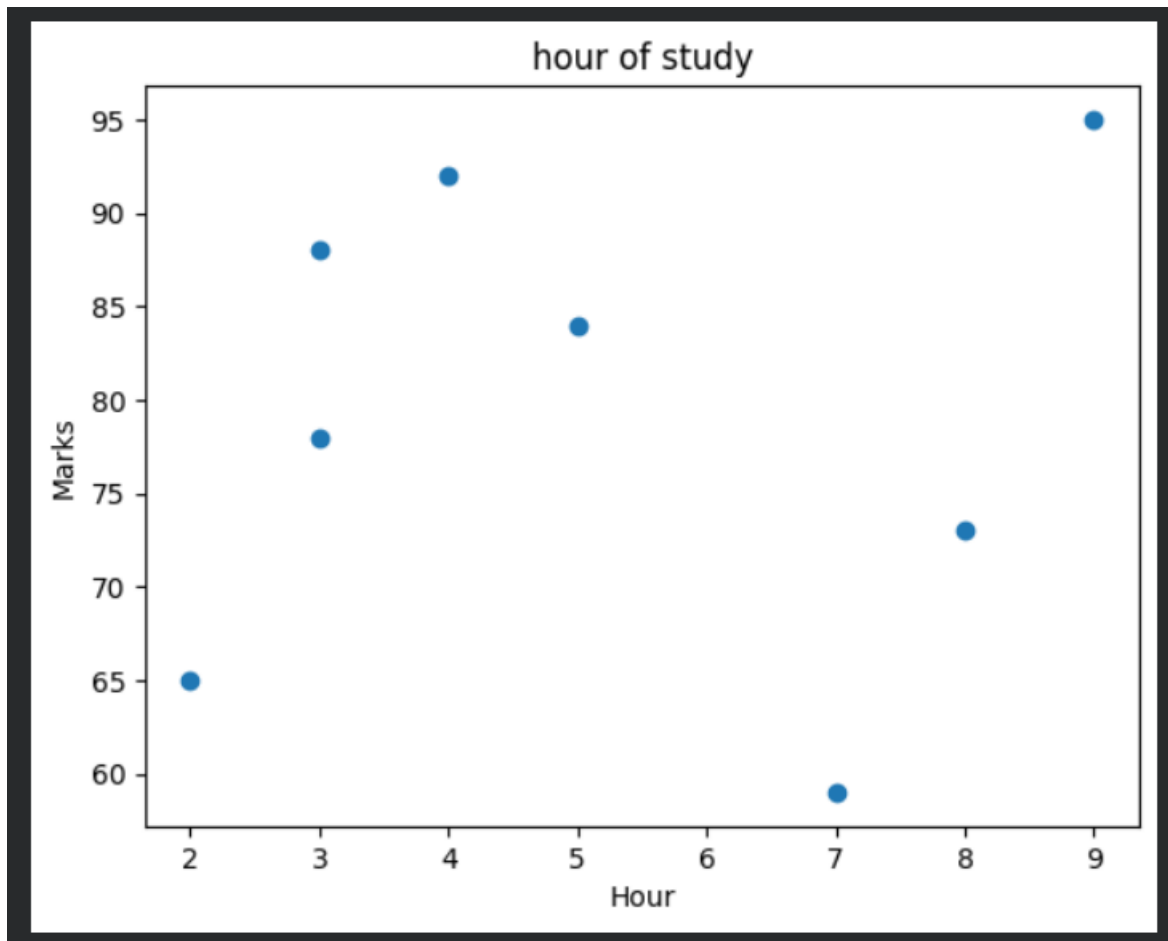
```
import matplotlib.pyplot as plt
marks = [78,92,65,84,59,73,88,95,41,67,82,90,55,76,98,61,70,85,93,48,79,87,64,91,58,74,83,96,69,80]
plt.hist(marks,bins=5)
plt.show()
```



11.

11.create a scatter plot showing study hours vs marks

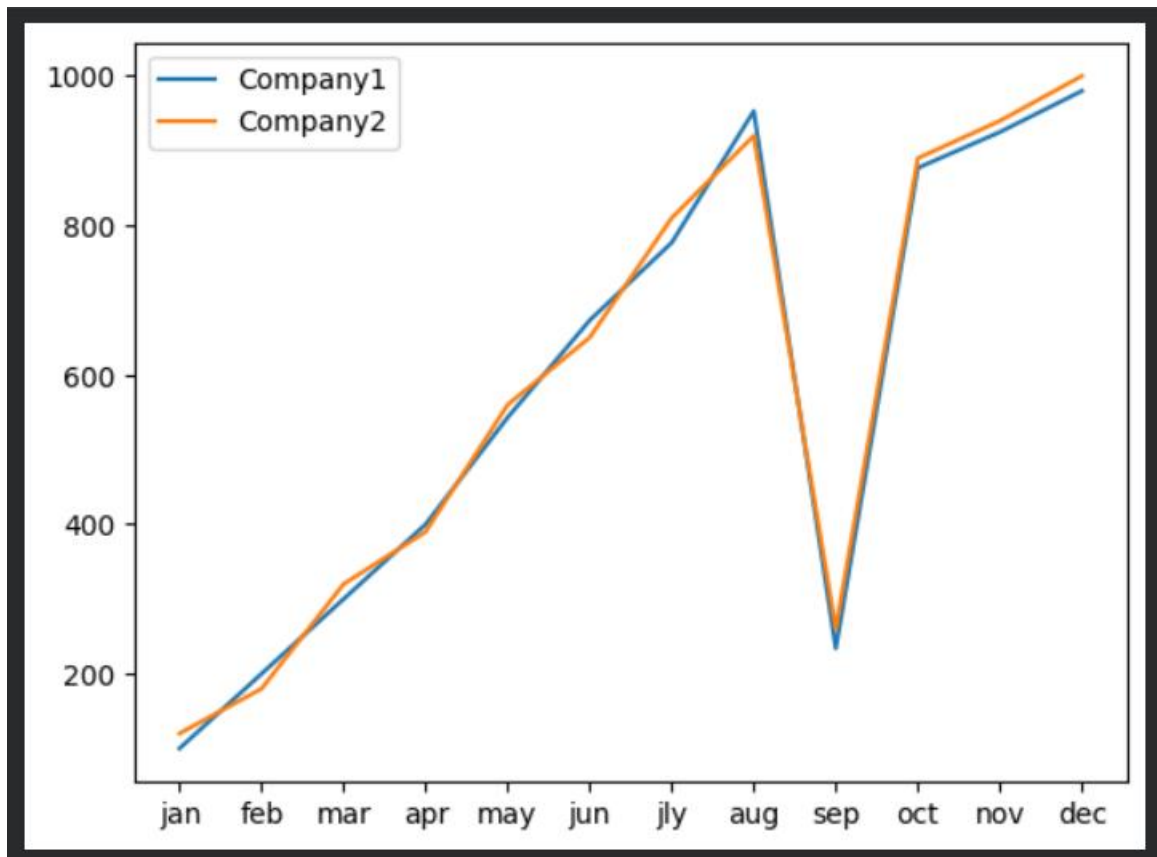
```
import matplotlib.pyplot as plt
hour=[3,4,2,5,7,8,3,9]
marks = [78,92,65,84,59,73,88,95]
plt.scatter(hour,marks)
plt.title("hour of study")
plt.xlabel("Hour")
plt.ylabel("Marks")
plt.show()
```



12.

12.compare sales of two companies using two lines plots in graph

```
import matplotlib.pyplot as plt
month=["jan","feb","mar","apr","may","jun","jly","aug","sep","oct","nov","dec"]
company1=[100,200,300,400,543,673,777,953,234,877,925,980]
company2=[120,180,320,390,560,650,810,920,260,890,940,1000]
plt.plot(month,company1,label="Company1")
plt.plot(month,company2,label="Company2")
plt.legend()
plt.show()
```



13.

13.create two subplots first barchart, second piechart

```
import matplotlib.pyplot as plt
plt.subplot(1,2,1)
plt.bar([1,2,3],[4,5,6])
plt.title("Barchart")
plt.subplot(1,2,2)
plt.pie([5,8,7],labels=["A","B","C"],autopct="%1.1f%%")
plt.title("Piechart")
plt.show()
```

